

Other Viruses II and Prion



2201 321 Veterinary Virology

Wilasinee Surasa, DVM, M.Sc.
Lecturer, Faculty of Veterinary Sciences
Mahasarakham University



วัตถุประสงค์

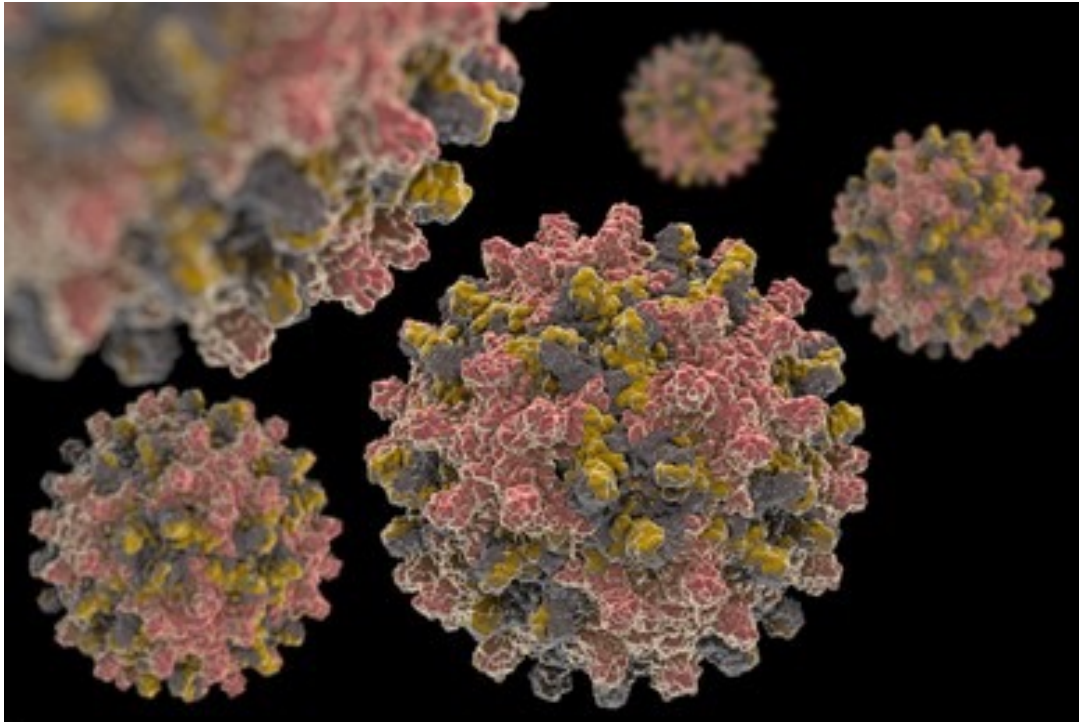
- นิสิตมีความรู้ความเข้าใจเกี่ยวกับลักษณะและชีววิทยาของไวรัส การก่อโรคในสัตว์ พยาธิกำเนิดและอาการแสดงทางคลินิก ตัวอย่างเพื่อชั้นสูตรทางห้องปฏิบัติการ วิธีการ และเทคนิคที่ใช้ ชั้นสูตรทางห้องปฏิบัติการ การควบคุมและการป้องกันโรคที่เกิดจากไวรัสต่างๆ ดังนี้
 - Hepeviruses
 - Hepatitis B virus
 - Hepatitis D virus
 - Anelloviruses
 - Prion

Hepeviruses

- Non-enveloped RNA virus
- Rat hepevirus was identified in ferrets in the Netherlands
- Novel hepevirus strains from African, Central American, and European bats are included in the species orthohepevirus D
- There is no evidence to date of transmission of the bat hepeviruses to humans
- Antibodies to hepeviruses have been reported in several other animal species including dogs, cats, goats, sheep, cattle, and non-human primates, however, the viruses potentially responsible for these infections are yet to be described



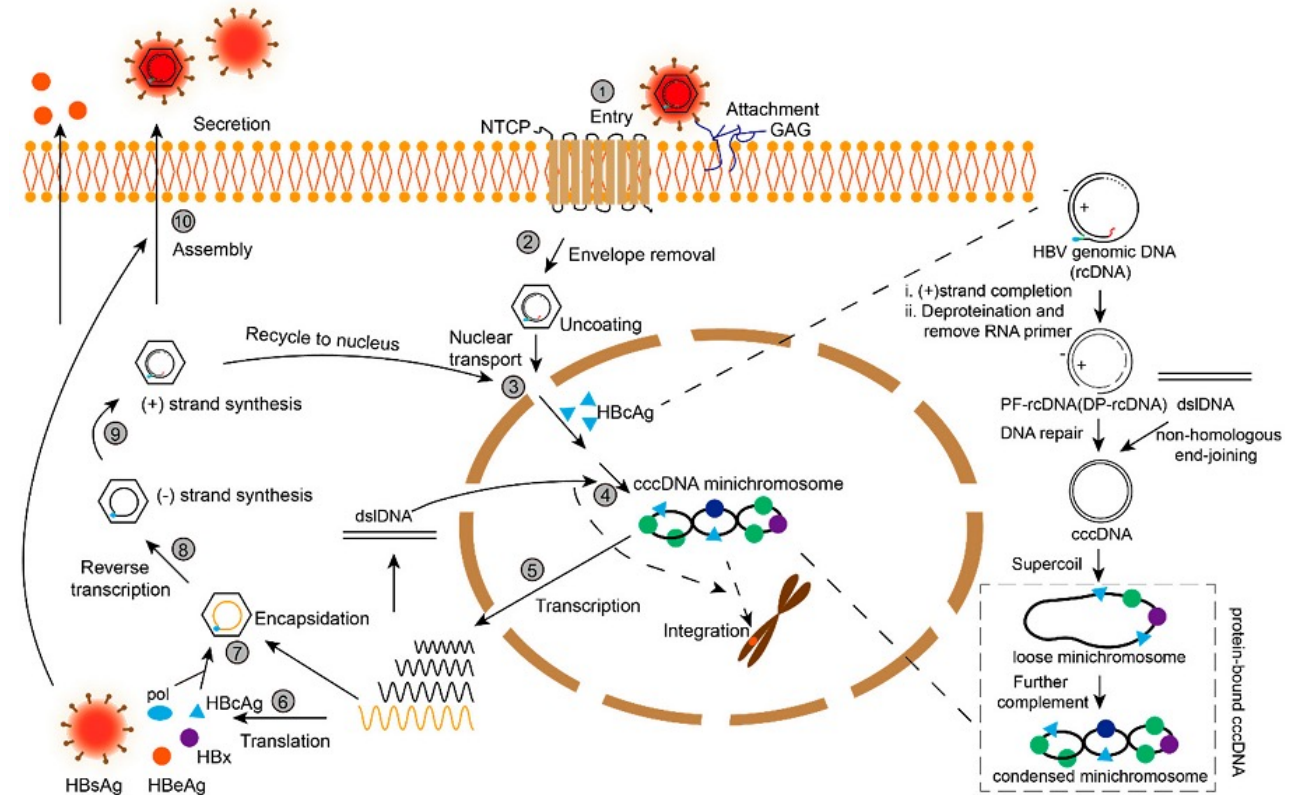
Family *Hepadnaviridae*



- Virion: envelope of 42–50 nm diameter surrounding an icosahedral nucleocapsid usually composed of 240 protein subunits
- Genome: 3.0–3.4 kb partially double-stranded DNA (circular)
- Host range: Teleost fish (*Parahepadnavirus* and *Metahepadnavirus*), reptiles and frogs (*Herpetohepadnavirus*), birds (*Avihepadnavirus*), mammals (*Orthohepadnavirus*)

Hepadnavirus replication cycle

- **Baltimore classification:** Class VII virus
 - Viral attachment and entry to hepatocyte via glycosaminoglycan (GAG) and sodium taurocholate cotransporting polypeptide (NTCP)
 - Removal of the envelope and the inner nucleocapsid is released into the cytoplasm
 - Transport of nucleocapsid into the nucleus pore and release of the polymerase (pol)-linked relaxed circular DNA (rcDNA)
 - Conversion of cccDNA from rcDNA and double-stranded linear DNA (dsDNA)
 - Transcription of viral mRNAs from cccDNA minichromosomes
 - Translation of viral proteins
 - Encapsidation involving pgRNA, pol and HBcAg
 - Synthesis of minus strand via reverse transcription of pgRNA → plus strand synthesis
 - Assembly of capsid containing rcDNA with antigen to form virion which secrets out of the cell

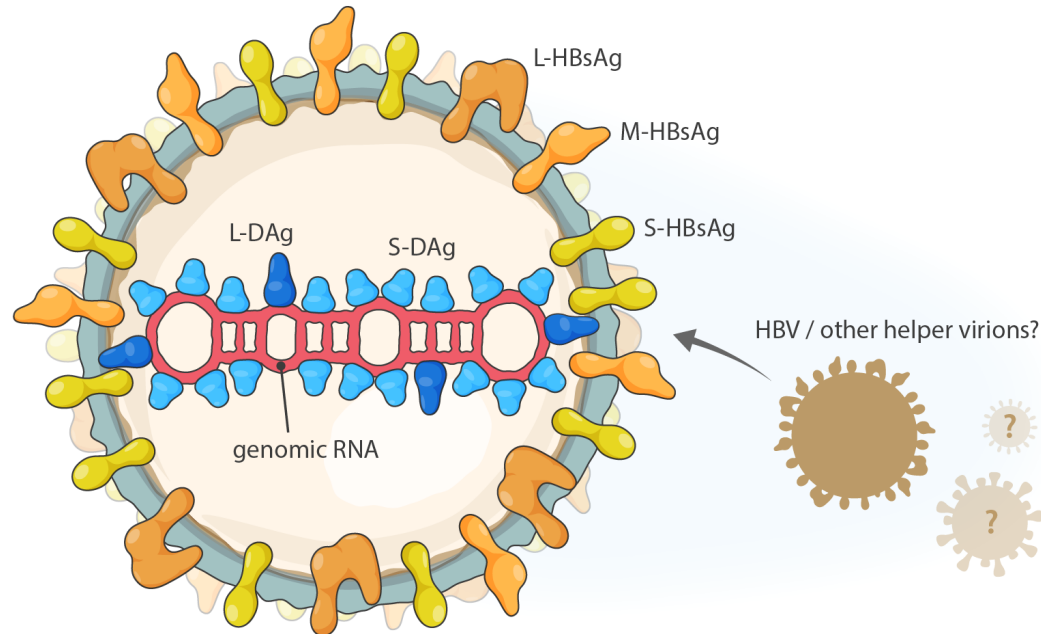


Dashed line: Variable 3'-end of the plus-strand; Red line: RNA primer at the 5'-end of the plus-strand; Green line: Redundant sequences at 5'- and 3'-end of the minus-strand; Blue oval: Polymerase

Hepatitis B virus (HBV)

- Spherical, or pleomorphic shaped
- In human: Chronic hepatitis B virus (HBV) infection increases the risk of developing liver fibrosis, cirrhosis and hepatocellular carcinoma (HCC), and it is a leading cause of mortality around the world
 - There are approximately 240 million chronic hepatitis B (CHB) patients worldwide and 680,000 deaths due to CHB annually
- Hepadnavirus infections of animals: duck (DHBV) and heron hepatitis B viruses, ground squirrel hepatitis virus
 - Recently, a novel hepadnavirus, similar to HBV, was identified in domestic cats and dog; hepatocellular carcinoma
- Diagnosis: commercial test kit, PCR
- Prevention: recombinant DNA vaccine

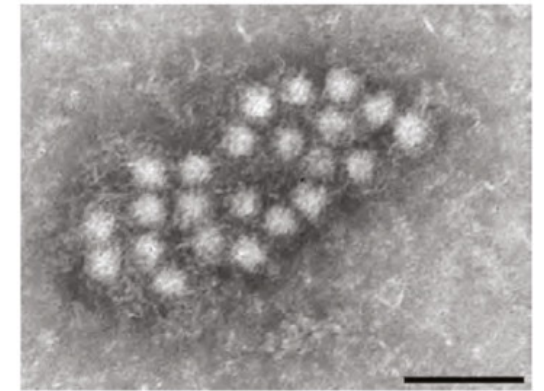
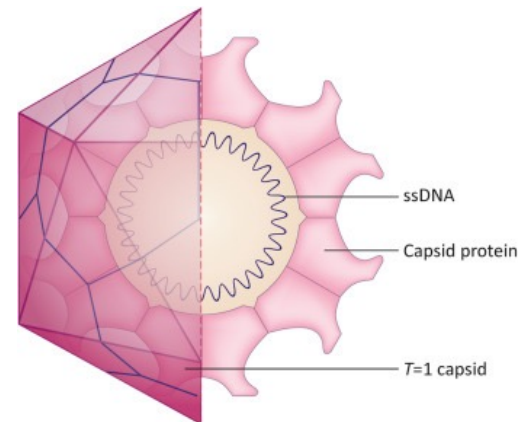
Family *Kolmioviridae*; Genus *Deltavirus*



- Hepatitis D virus: a satellite virus (the primary helper virus of HDV; hepatitis B virus (HBV)
 - In human: Coinfection with HBV and HDV can cause serious short-term health problems and even liver failure and death
- Deltavirions are spherical (36–43 nm in diameter) with a lipid envelope containing peplomers obtained from hepatitis B virus (HBV, family *Hepadnaviridae*)
- Virions contain a ribonucleoprotein (RNP) complex consisting of about 70 copies of small delta antigen (S-DAG) and large delta antigen (L-DAG)
- Genomes: non-segmented, negative-sense, covalently-closed circular RNA (cccRNA) of about 1.7 kb

Anelloviruses

- Non-enveloped, icosahedral symmetry, with a diameter of about 30-32 nm
- Genome: circular ssDNA genome (-) genome of about 3.8 kb
- Virus replication: direct penetration → uncoating → in nucleus; viral ssDNA genome was converted to dsDNA → mRNA → translate viral proteins → ssDNA synthesis from dsDNA template → release by cell lysis
- Natural hosts: Human, Chimpanzee, African monkey, Tupaia, Pig, Cow, Sheep and Chicken
- Transmission: sexual, blood, possibly also fecal-oral
- Symptoms: Asymptomatic
 - May be associated with various diseases: hepatitis, pulmonary diseases, hematologic disorders, myopathy and lupus
- No specific vaccine and antiviral drug

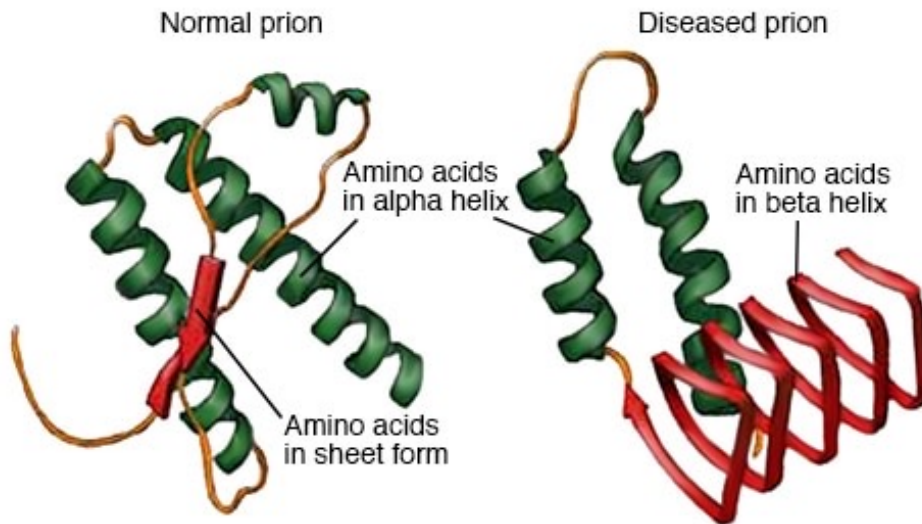




Prion disease or Transmissible spongiform encephalopathies (TSEs)

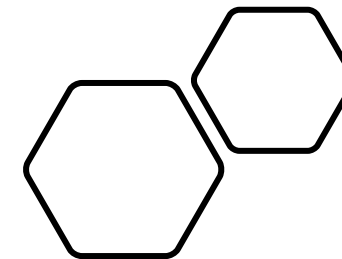
- **Neurodegenerative disorders**
 - มีพยาธิสภาพเกิดขึ้นที่สมองและมีอาการแสดงของระบบประสาททั้งในคนและในสัตว์
- They are distinguishing by long incubation period with characteristic of spongiform changes associate with neuronal loss
- **Prion** – Small **proteinaceous** **infectious** particle (30-50 nm)

Prion

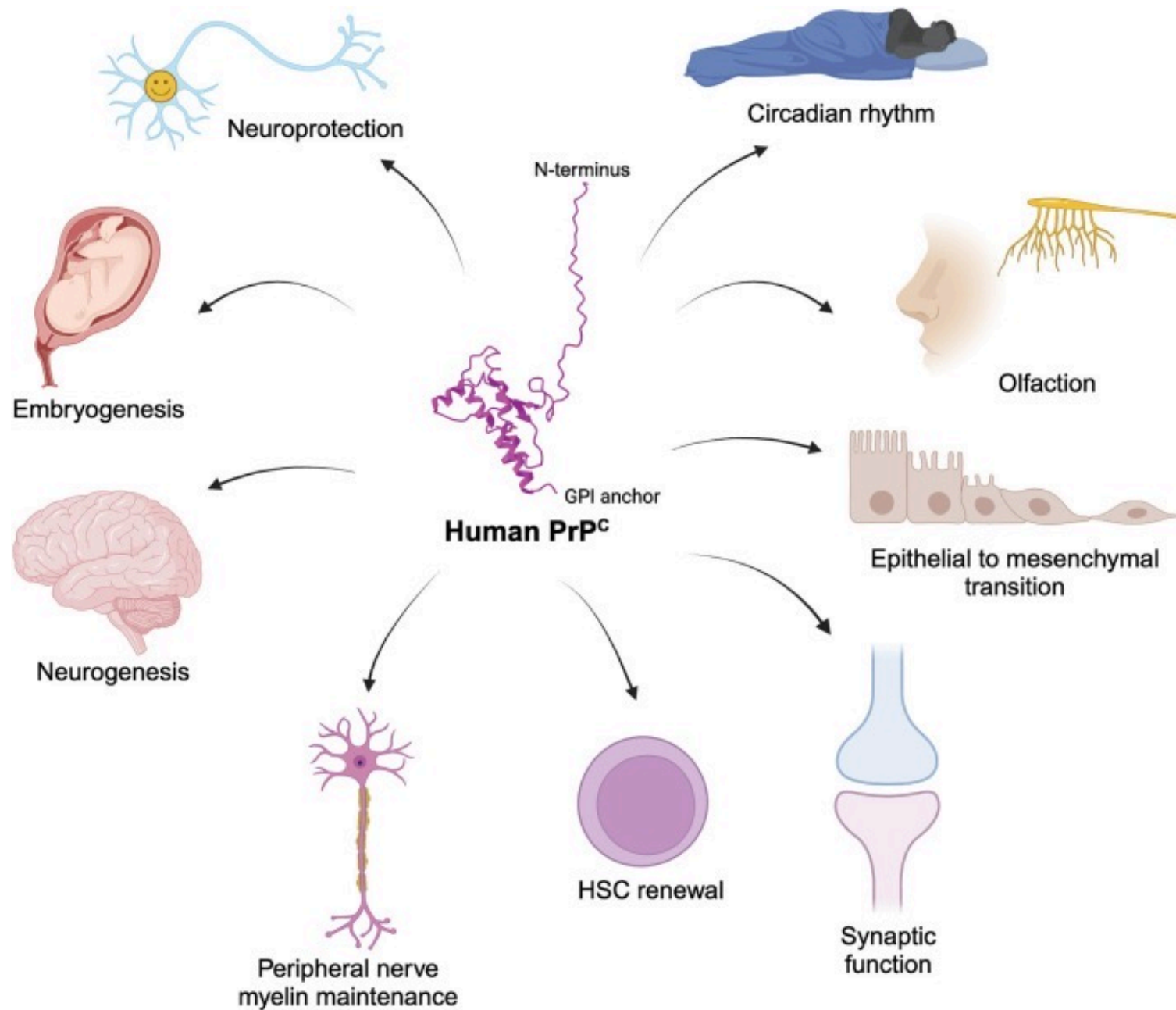


© MAYO FOUNDATION FOR MEDICAL EDUCATION AND RESEARCH. ALL RIGHTS RESERVED

- **PrP^C (normal protein)** – prion found throughout the body of healthy people and animals (cell etc. brain, T- or B-lymphocytes)
 - Maintenance of long-term memory
 - Immunoregulation
 - Signal transduction
 - Copper binding
 - Synaptic transmission
 - Induction of apoptosis or protection against apoptotic stimuli
- **PrP^{Sc} (infectious protein) – scrapie form**
- **Resistant to:**
 - High temperature
 - High pressure
 - Formaldehyde, phenol, chloroform
 - Protease, nuclease
 - UV-irradiation



The importance of prion



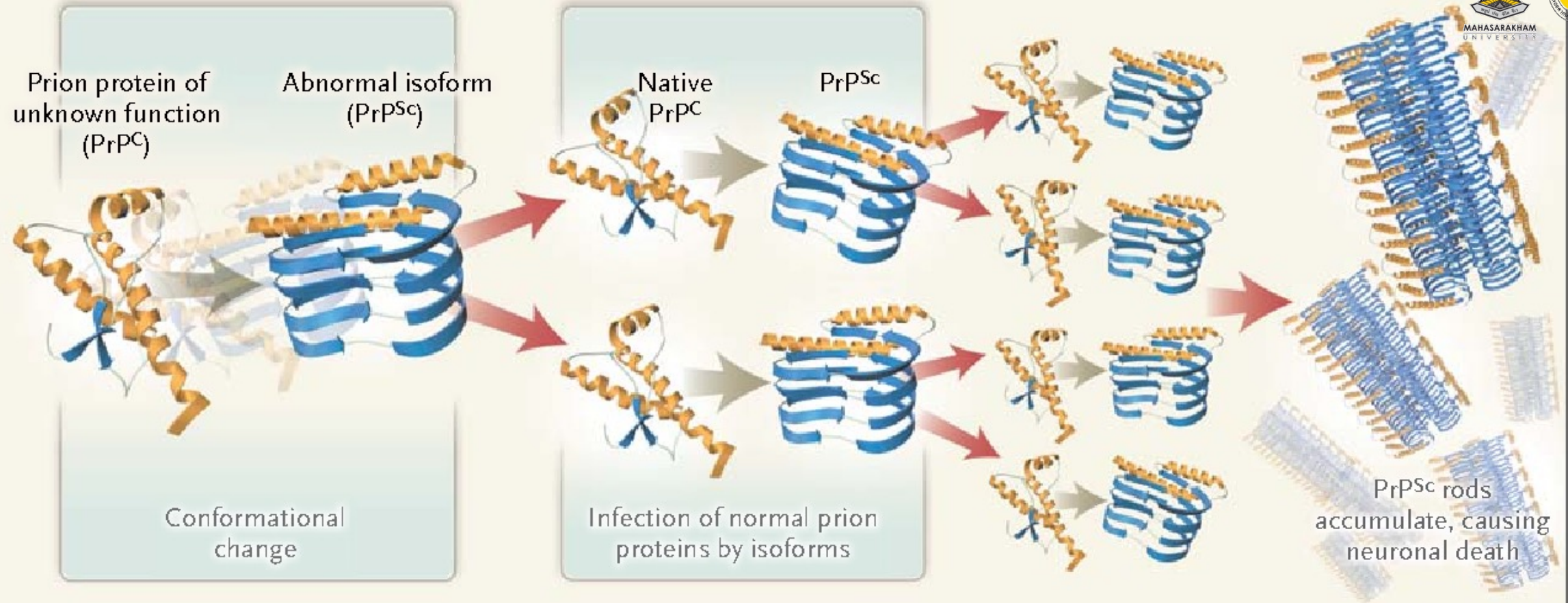


Figure 4. Conversion of Normal Prion Proteins to Abnormal Isoforms.

In prion diseases, the prion protein (PrP^C) undergoes a conformational change in which it becomes an abnormal, pathogenic isoform (PrP^{Sc}). This abnormal isoform can then induce the same conformational change in other native prion proteins, leading to an accumulation of abnormally folded aggregates (PrP^{Sc} rods) that lead to neuronal death.

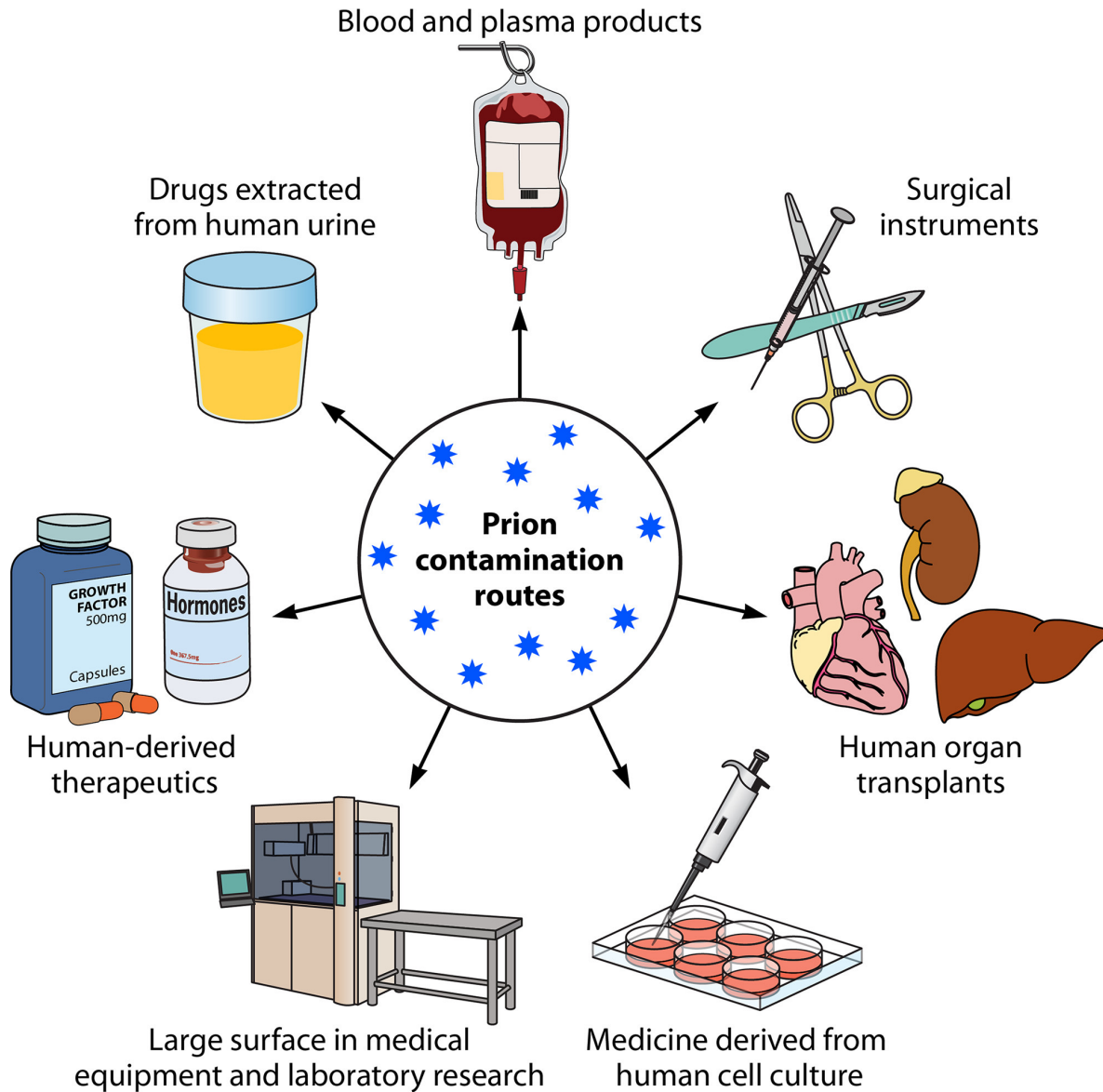
Prion associated disease

- Cause prion disease
 - CNS involvement
 - Capacity of infected brain tissue to transmit disease
 - Abnormal accumulation of abnormal protein
 - Absence of immune response
 - Long incubation period (months to years)
 - Lead to fatal



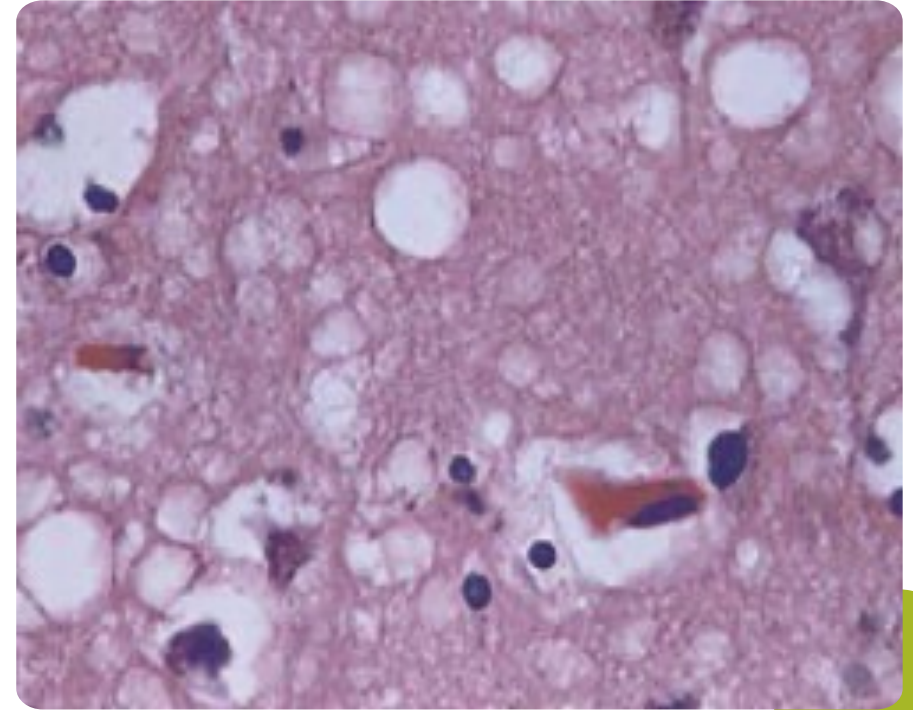
Transmission

- Ingestion
- Familial (genetic)
- Brain inoculation
- Contamination route

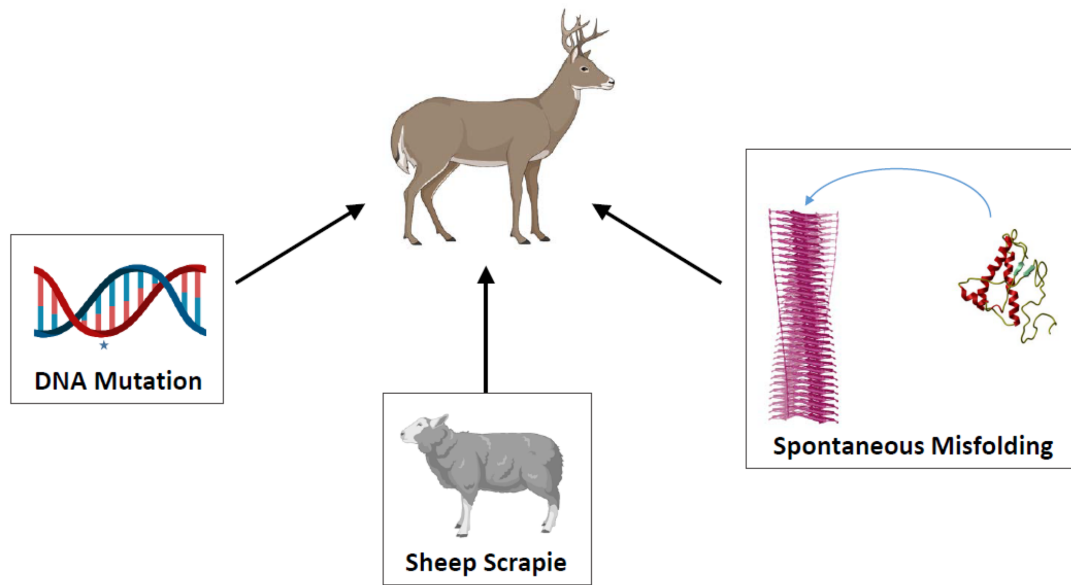


Transmissible spongiform encephalopathies (TSE)

- Because after the infection, the brain appearance become a sponge-like appearance
 - **T:** Transmissible (communicable)
 - **S:** Spongiform (look like sponge)
 - **E:** Encephalopathy (brain diseases)



In animals:

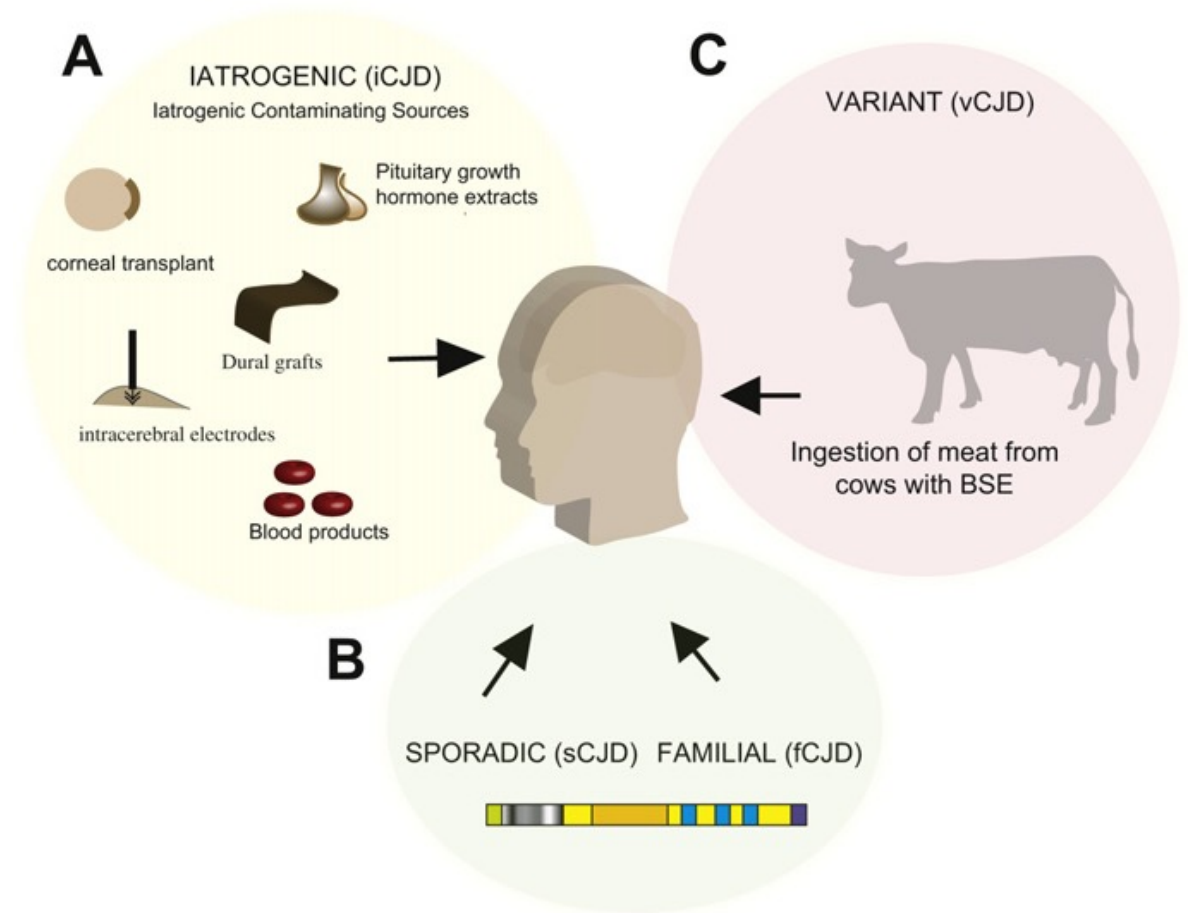


Transmission of Chronic wasting disease (CWD)

- Scrapie in sheep and goats
- Bovine spongiform encephalopathies (BSE) in cattle - Mad cow disease
- Transmissible mink encephalopathies (TME) in mink
- Chronic wasting disease (CWD) in elk and mule deer
- Feline spongiform encephalopathy in cat
- Exotic ungulate encephalopathy (EUE) in nyala, oryx and greater kudu
- Spongiform encephalopathy of the ostrich

In human:

- Creutzfeldt-Jakob disease (CJD) and its varieties:
 - Iatrogenic Creutzfeldt-Jakob disease (iCJD)
 - Variant Creutzfeldt-Jakob disease (vCJD)
 - Familial Creutzfeldt-Jakob disease (fCJD)
 - Sporadic Creutzfeldt-Jakob disease (sCJD)
- Gerstmann-Steausler-Scheinkersyndrom (GSS)
- Fatal familial insomnia (FFI)
- Sporadic fatal insomnia (SFI)
- Kuru



Pathological features of prion diseases

- The main pathological features of prion diseases –
 - **Spongiform change:** the most characteristic feature, where small vacuoles (holes) appear in the brain's gray matter, giving it a sponge-like appearance under a microscope
 - **Neuronal loss:** the presence of prions leads to a progressive death of nerve cells (neurons)
 - **Gliosis (astro- and microgliosis):** the brain responds to the damage by increasing the number of glial cells, specifically astrocytes and microglia, in a process known as gliosis
 - **PrP^{Sc} deposits:** abnormal prion protein (PrP^{Sc}) aggregates and accumulates in the brain
 - These deposits can range from small, diffusely distributed molecules to larger, more organized structures like plaques, depending on the specific disease type
 - **Amyloid plaques:** some prion diseases are characterized by the formation of amyloid plaques, which are deposits of misfolded proteins
 - In certain forms, such as Gerstmann-Sträussler-Scheinker (GSS) disease, amyloid plaques are a defining feature



Scrapie

- Scrapie has been known since the 18th century (1732) and does not appear to be transmissible to humans
- Scrapie is fatal, degenerative disease affecting CNS of sheep and goats, incubation period in sheep is 2-5 years
 - Signs vary and develop very slowly
 - Early signs include subtle changes in behavior or temperament
 - These changes may be followed by scratching and rubbing against fixed objects, apparently to relieve itching
- Due to damage to nerve cells (CNS), affected animals usually show behavioral changes, excitability, tremor (especially of head and neck), grinding of teeth, pruritus, and locomotor incoordination that progresses to recumbency and death within 6 weeks – 6 months

Transmission

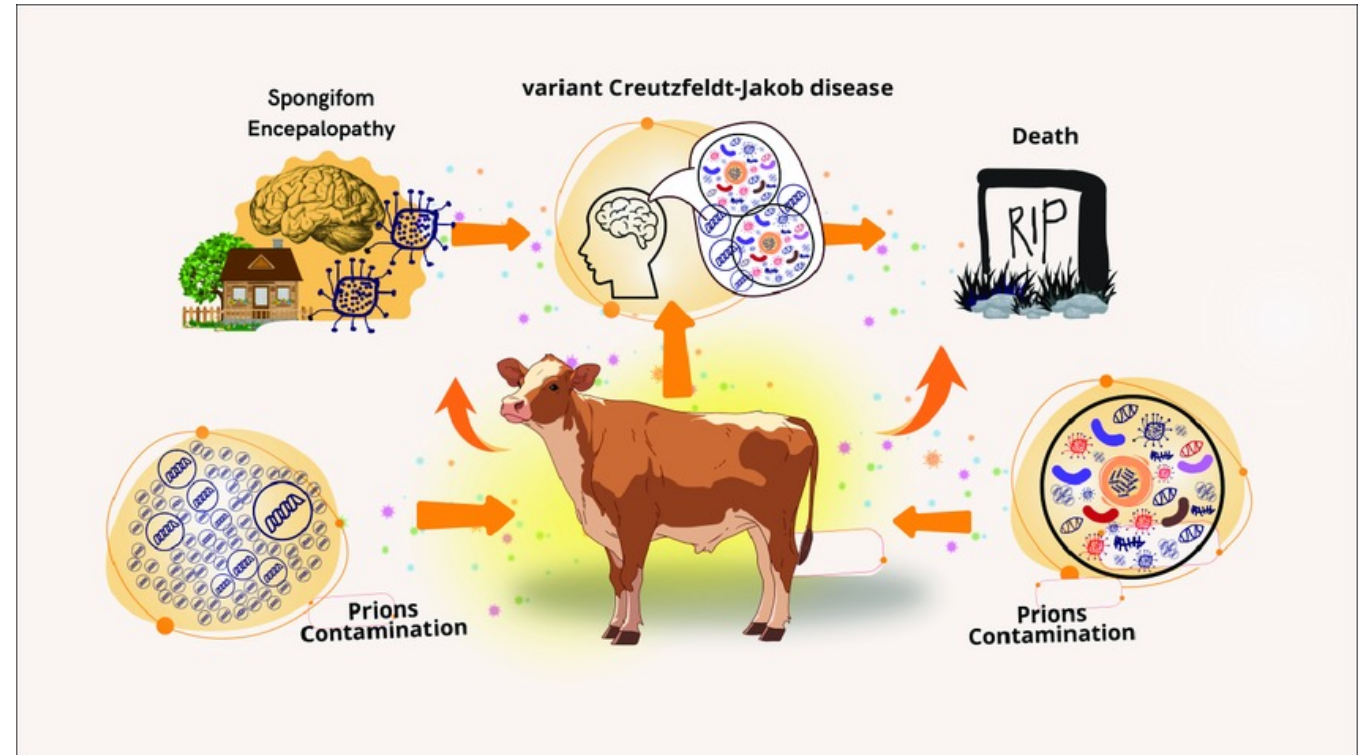
- Oral route
- Superficial wounds
- In experiment: peripheral route (IP, SC, IV)
- Prolonged incubation periods
- Intra-cerebral inoculation
- Short incubation periods
- Transplacental - ???

Bovine spongiform encephalopathies (BSE)

- Caused by contamination of meat and bone meal with a scrapie-like agent
- BSE has a long incubation period ~ 4 years (2-15), usually affecting adult cattle (age onset of 4-5 years), all breeds being equally susceptible
- **Clinical signs:** CNS dysfunction, tremors, behavior changes, hypersensitivity to various stimuli, hyperesthesia with kicking during milking, abnormal posture, hindlimb ataxia, aggression, reduced milk yield, weight loss death within 6 months
- Transmitted by oral or intracerebral route
- Believed but not proven to transmit to human via eating infected carcass

BSE; way to infection

- Cow eat offal of the infected sheep
- Prions taken up from the gut and transported along nerve fibers to the brain stem
- Prions accumulate and convert normal prion proteins to the disease-causing form PrP^{Sc}
- Years later, BSE results when a sufficient number of nerve cells have become damaged, affecting the behavior of the cows and eventually the cow is dead



References

- Carter, J.B. and Saunders, V.A. 2007. Virology: Principles and Applications. John Wiley & Sons Ltd., England. 358p.
- Hirsh, D.C., MacLachlan, N.J. and Walker, R.L. 2004. Veterinary Microbiology. 2nd edition. Blackwell Publishing. Ames, Iowa. 536p.
- MacLachlan, N.J. and Dubovi, E.J. 2011. Fenner's Veterinary Virology. 4th edition. Elsevier Inc., UK. 507p.
- Mahy, B.W.J. and Van Regenmortel, M.H.V. 2010. Desk Encyclopedia of Animal and Bacterial Virology. Elsevier Inc., UK. 617p.
- Quinn, P.J., Markey, B.K., Leonard, F.C., FitzPatrick, E.S., Fanning, S. and Hartigan, P.J. 2011. Veterinary Microbiology and Microbial Disease. 2nd edition. Wiley-Blackwell, West Sussex, UK. 912p.



E-mail address: wilasinee.c@msu.ac.th